StudyMapper

AI-Driven Personalized Learning and Academic Management System

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Agenda

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Abstract & Motivation

Abstract

StudyMapper delivers AI-driven personalized learning through adaptive study paths, intelligent content generation (notes, quizzes, diagrams), and comprehensive analytics for students and educators.

Motivation: Rising educational challenges, need for personalization, limited adaptive assessment, demand for integrated academic management.

Key Innovations

- SM-2 Spaced Repetition
- Adaptive Difficulty Planner
- AI Content Generation (OpenRouter)
- Weakness Detection (0-100)
- Gamification Engine
- 40 DB Tables | 47/47 Tests
- Production Build | Live Demo

Problem Statement

Current Educational Challenges

- One-size-fits-all teaching approach
- Lack of personalized learning paths
- Poor student engagement and motivation
- Difficulty tracking individual progress
- Limited adaptive assessment methods
- Inefficient academic resource management
- Rising academic stress and dropout concerns
- Need for data-driven educational interventions

Objectives & Scope

Objectives
- Develop adaptive personalized learning
- Implement AI content generation
- Create real-time analytics
- Design intelligent study planner
- Build comprehensive management
- Ensure privacy/security
- Scalable infrastructure

Scope
- Higher education (undergraduate)
- Computer Science & Engineering
- Multiple subjects (DBMS/OS/DSA)
- Adaptive paths | Gamified experience
- Analytics dashboard | Mobile responsive

Literature Survey

Adaptive Learning
- MIT Open Learning
- Khan Academy paths
- Coursera recommendations
- Adaptive tutoring systems

AI in Education
- OpenAI educational
- Google AI for Teaching
- IBM Watson Education

Learning Management
- Moodle analytics
- Blackboard adaptive
- Canvas tutoring

Spaced Repetition
- Anki SM-2
- Quizlet adaptive
- SuperMemo systems

Proposed System

Core Modules
- Learning Management Platform
- AI Content Generator (notes, quizzes, diagrams)
- Adaptive Study Planner (SM-2, panic mode)
- Performance Analytics Dashboard
- Gamification Engine (XP, levels, streaks, achievements)
- Smart Flashcards with scheduling
- Placement Preparation Module
- Coding Practice (pyodide environment)
- AI Study Assistant (OpenRouter 6-model)
- Mobile Responsive Design

System Architecture

Frontend (React 19 + Vite)
- Components & Pages
- Context (Auth, Theme)
- 46-Route Router
- Responsive Layout

Backend Services
- Firebase Auth
- Dexie.js IndexedDB
- File Parser (Tesseract, pdf.js, mammoth)
- OpenRouter Integration

AI & Analytics Layer
- OpenRouter (6 free models)
- Streaming + JSON Repair
- SM-2 Spaced Repetition
- Weakness Detector (0-100)
- Adaptive Planner
- Mermaid Diagrams (KaTeX)

Methodology & Workflow

Phase 1: Analysis
- Problem identification
- Requirements gathering
- Domain research

Phase 2: Design
- System architecture
- Database schema (40 tables)
- API design

Phase 3: Implementation
- Component development
- Integration testing
- Performance optimization

Phase 4: Validation
- 47/47 tests passing
- Build verification
- Demo data seeding

Technology Stack

Frontend
- React 19
- Vite
- JavaScript
- CSS Modules
- Responsive Design

Backend / Auth
- Firebase
- Dexie.js (IndexedDB)
- Google/Email/Phone Auth
- Sync Capability

AI / Tools
- OpenRouter API
- Pyodide
- Tesseract OCR
- pdf.js / mammoth
- Mermaid / KaTeX
- SM-2 Algorithm

Database Design & Data Flow

Schema (40 Tables)
- subjects, units, topics
- topic_contents, flashcards
- study_plans, gamification_profiles
- placement_*, coding_problems
- ai_*, study_analytics
- weak_score, question_bank

Data Flow
- IndexedDB (primary storage)
- Firebase sync (optional)
- File parsing pipeline
- AI content generation
- Analytics computation
- Offline-first design

UI Design & Key Features

Verified Features (Real Screenshots from Running App)
- Dashboard with analytics and progress tracking
- Subject view (DBMS, OS, DSA) with topic content
- AI-generated topic notes with Mermaid diagrams
- Study planner (scheduled and empty states)
- Skill roadmap (browse and detail views)
- Placement prep, question bank, mock exam setup
- Coding practice workspace (pyodide environment)
- Flashcards with SM-2 scheduling interface
- AI assistant dashboard (6-model fallback)

Results & Experimental Analysis

Testing Results
- 47/47 Tests Passed (100%)
- Build Time: 6.21s
- Zero Critical Errors
- All Components Functional

Demo Data
- 38 Topics across 3 subjects
- 18 Topic Contents (rich notes)
- 8 DBMS Questions + AI answers
- Mock Exams with scoring
- Flashcards with SM-2 fields

Performance Metrics
- Production build verified
- Live demo available
- Responsive design tested
- Offline functionality confirmed
- File parsing verified
- AI streaming working
- JSON repair mechanism verified

Discussion & Contributions

Contributions
- Adaptive learning engine (SM-2 algorithm)
- AI content pipeline (notes, quizzes, diagrams)
- Gamification system (XP, levels, streaks)
- Weakness detection scoring (0-100)
- Panic mode intervention

Limitations
- Firebase/OpenRouter keys not exposed
- Some supervisor details pending
- Mobile app development future work
- Expanded multi-language planned

Conclusion

StudyMapper Successfully Implemented

- AI-driven personalized learning platform
- Adaptive study planner with SM-2 spaced repetition
- AI content generation via OpenRouter (6-model fallback)
- Comprehensive analytics and weakness detection
- Gamified experience with achievements and quests
- Production-ready: 47/47 passing tests
- Live demo at study-mapper.vercel.app

Guide: Ms.C. Sravani | Department: Computer Science and Engineering
Institution: Anantha Lakshmi Institute of Technology and Sciences (Autonomous)

Future Scope

Short-term
- Enhanced AI model selection
- Expanded content library
- Improved analytics visualization
- Mobile app development
- Collaboration features
- Better file parsing

Long-term
- Multi-language content
- Full offline synchronization
- Institutional deployment
- Integration with LMS platforms
- Research collaboration
- Expanded AI capabilities

References

Verified Sources

Project: GitHub palturuvenubhaskar/StudyMapper | Abstract PDF (ALITS)
Framework: React 19 (react.dev) | Vite (vitejs.dev) | Dexie.js (dexie.org) | Firebase (firebase.google.com)
AI: OpenRouter API (openrouter.ai) | Pyodide (pyodide.org)
Parsing: Tesseract.js | pdf.js (mozilla) | mammoth.js | Mermaid.js | KaTeX
Algorithm: SM-2 Spaced Repetition (SuperMemo/Anki implementation)

Thank You

Questions & Discussion

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Live Demo: study-mapper.vercel.app | GitHub: palturuvenubhaskar/StudyMapper
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